



TCET
DEPARTMENT OF COMPUTER ENGINEERING
Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education
Under Autonomy - CBCGS-HME 2023
University of Mumbai



Software Requirements Specification

for

Shopkart E-commerce website

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this project is to develop an E-Commerce System that allows customers to browse products, place orders, and make online payments through a web-based platform. The system also provides an admin panel for administrators to manage product inventory, orders, and user accounts. The project aims to provide a secure, efficient, and user-friendly online shopping platform.

1.2 Document Conventions

N/A

1.3 Intended Audience and Reading Suggestions

The main users of the E-Commerce System are customers, who browse products, add items to the cart, and purchase products online, and administrators, who manage the product inventory, update product details, monitor orders, and control the overall system operations.

1.4 Product Scope

The e-commerce website enables users to browse products, manage carts, process payments, and track orders. It supports B2C transactions, integrates with payment gateways like razorpay. Future enhancements may include features like AI recommendations, AR virtual try-on, blockchain authenticity verification, NFT loyalty rewards, and voice search, mobile apps etc.

1.5 References

N/A

2. Overall Description

2.1 Product Perspective

The E-Commerce System is a web-based application developed to provide an online platform for buying and selling products. It acts as an independent system that allows customers to browse products, place orders, and make payments online, while administrators manage product inventory, orders, and user accounts through an admin panel. The system works over the internet and integrates with databases and payment gateways to provide real-time product and order management.

2.2 Product Functions

Core functions include product browsing with grid/list views and image zoom, advanced search with filters, guest and registered checkout, wishlist with one-click social sharing, order tracking, and personalized dashboards. Additional capabilities support stock alerts, and review submissions.

2.3 User Classes and Characteristics

- Customers: Everyday shoppers aged 18-55 using desktops or mobiles; prefer fast, intuitive interfaces with guest checkout.

- Guests: Casual browsers who avoid registration but use quick features like zoom and sharing.

2.4 Operating Environment

- Web Browser(Chrome, Firefox, Edge).
- Android and IOS Devices
- Server running on Windows
- Internet connection required

2.5 Design and Implementation Constraints

- Follows Tailwind CSS framework, REST APIs with JSON.
- Secure payment gateway integration is mandatory
- System must support high traffic during peak hours
- PostgreSQL database must be used
- Must comply with data privacy laws

2.6 User Documentation

N/A

2.7 Assumptions and Dependencies

N/A

3. External Interface Requirements

3.1 User Interfaces

The e-commerce website shall provide a responsive and easy-to-use web interface accessible on desktops, tablets, and mobile devices. It includes pages for user registration and login, product browsing, shopping cart, checkout, order confirmation, and user account management. An admin interface is provided for managing products, orders, and users.

3.2 Hardware Interfaces

This outlines the hardware specifications needed for the software to function, such as memory limits, cache size, processor, RAM capacity, etc.

1. Intel/Ryzen Processor
2. SSD 256 GB
3. RAM 8 GB
4. Cache 1 MB.

3.3 Software Interfaces

The website interfaces with web browsers, a database system for storing data, payment gateway software for processing transactions, and email services for notifications. It may also integrate with third-party APIs for shipping, analytics, and reporting.

3.4 Communications Interfaces

NA

4. System Features

4.1 User Authentication and Login System

Description and Priority

Provides user registration, login, password recovery, and dark/light mode interface customization.

Priority: High

Functional Requirements

REQ-4.1.1: The system shall allow users to create accounts and log in securely.

REQ-4.1.2: The system shall support password recovery.

REQ-4.1.3: The system shall store credentials securely.

REQ-4.1.4: The system shall provide dark and light display modes.

4.2 Product Browsing and Search System

Description and Priority

Enables users to browse, search, filter, and view detailed product information.

Priority: High

Functional Requirements

REQ-4.2.1: The system shall display categorized product listings.

REQ-4.2.2: The system shall support keyword search and filters.

REQ-4.2.3: The system shall display product details and images.

4.3 Online Payment Processing System

Description and Priority

Supports secure online payments using integrated payment gateways.

Priority: High

Functional Requirements

REQ-4.3.1: The system shall support COD payment method.

REQ-4.3.2: The system shall confirm transaction status.

4.4 Order Tracking and User Account Management

Description and Priority

Allows users to manage profiles and track order history and status.

Priority: Medium

Functional Requirements

REQ-4.4.1: The system shall maintain user order records.

REQ-4.4.2: The system shall display order tracking information.

4.5 Admin Product and Order Management System

Description and Priority

Enables administrators to manage products, inventory, and customer orders.

Priority: High

Functional Requirements

REQ-4.5.1: The system shall allow admins to add, edit, and delete products.

REQ-4.5.2: The system shall update inventory levels.

REQ-4.5.3: The system shall manage order statuses.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

In order to maintain an acceptable speed at maximum number of uploads allowed from a particular customer as any number of users can access to the system at any time. Also the connections to the servers will be based on the attributes of the user like his location and server will be working 24X7 times.

5.2 Safety Requirements

Page loads should be returned and formatted in a timely fashion depending on the request being made.

Administrators will have the ability to edit the aspects of the order forms, product descriptions, prices and website directly

5.3 Security Requirements

Pages of the website must be access in the way they were intended to be accessed. Included files shall not be accessed outside of their parent file.

Administrator can only perform administrative task on pages they are privileged to access.

Customers will not be allowed to access the administrator pages

5.4 Software Quality Attributes

- **Usability:** The interface shall be responsive, automatically adjusting layout for Mobile and Desktop screens.
- **Availability:** The system shall be available 99.5% of the time during business hours
- **Reliability:** The shopping cart shall retain user items for 30 minutes even if the session is accidentally closed.
- **Maintainability:** The backend code shall be structured using Spring Boot standards to allow easy addition of new modules (e.g., Inventory Management).

5.5 Business Rules

- **BR-1:** Only users with the "Administrator" role can add, edit, or delete product listings.
- **BR-2:** Customers cannot view the Admin Dashboard.
- **BR-3:** Guest users must provide a valid email address to receive order receipts.
- **BR-4:** An order cannot be cancelled by the user once it is marked as "Shipped" by the admin.

6. Other Requirements

- **6.1 Database:** The system shall use MySQL or PostgreSQL for persistent data storage.

- **6.2 Legal:** The system shall display a "Terms of Service" and "Privacy Policy" link in the footer.
- **6.3 Currency:** All prices shall be displayed in the local currency (e.g., INR or USD) based on default settings.

Appendix A: Glossary

- **API (Application Programming Interface):** Software intermediary that allows two applications to talk to each other (e.g., Payment Gateway).
- **Guest User:** A visitor who purchases items without creating a permanent account.
- **PCI DSS:** Payment Card Industry Data Security Standard; security standards for organizations that handle credit cards.
- **UI (User Interface):** The visual part of the website users interact with.
- **B2C:** Business-to-Consumer transactions.

Appendix B: Analysis Models

N/A

Appendix C: To Be Determined List

TBD ID	Description	Target Resolution Date
TBD-01	Hosting Provider: The specific cloud provider (e.g., AWS Free Tier, Heroku, or Vercel) for deploying the live application is yet to be finalized.	By Design Phase End
TBD-02	Payment Gateway Mode: Confirmation on whether Razorpay will be implemented in "Test Mode" (Sandbox) or "Live Mode" for the demo.	By Implementation Phase
TBD-03	SMTP Server: The specific email service provider (e.g., Gmail SMTP or SendGrid) for sending order confirmations is pending selection.	By Implementation Phase
TBD-04	Logo & Branding: The final color palette and "Shopcart" logo assets are currently under design.	By UI/UX Phase

6.1.1 For Faculty Use Only

Correction Parameter	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
Marks Obtained				

